(Don't be) Bullied into Bad Science

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Slides: http://bit.ly/eglen-cost (CC-BY license)

Declarations

1. Member of *Bullied into Bad Science* campaign
2. Affiliate editor of *bioRxiv*
3. Senior editor of *Scientific Data*

Acknowledgements

1. Corina Logan and Laurent Gatto.
2. Joined by Ross Mounce, Adrian Curry, Lauren Maggio and myself.

http://bulliedintobadscience.org has most of the key background

Thanks to Ben Farrar and Yvonne Nobis for material.

What does it mean to be bullied into bad science?

1. Science is a competitive arena; long work hours culture
2. Race to publish first
3. Not being allowed to shared resources
4. Career and family often in competition.

Problems

We are evaluated on where, not what, we publish

Evaluation of research by impact factor or journal title

$$I_y = \frac{C_{y-1} + C_{y-2}}{P_{y-1} + P_{y-2}}$$

C_y = number of citations in year y, P_y = number of publications in year y...

How could this possibly go wrong?

Strong pressure e.g. in China to pressure in journals with impact factor over some threshold (5).

So many ways to game the system: *A user's guide to inflated and manipulated impact factors* (Ioannidis and Thombs, 2019).

Reproducibility crisis

Crisis? What crisis? Science is getting better, right?

"Negative results are disappearing from most disciplines and countries" (Fanelli, 2012). Study of ~4600 papers:

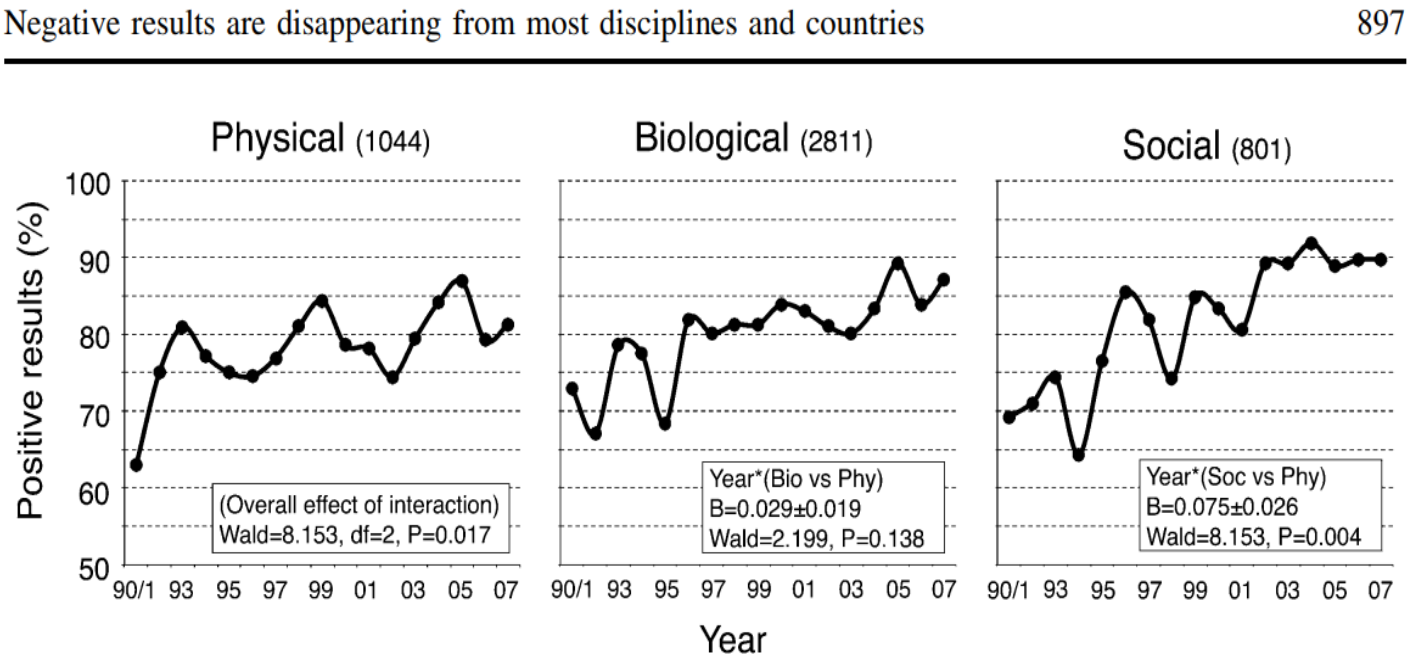


Fig. 3 Percentage of papers reporting a support for the tested hypothesis plotted against year of publication and divided by scientific domain of the journal (physical, biological and social sciences). Logistic regression estimates are interaction effects in a hierarchically well-formulated model. The main effects of this model, calculated with interaction components removed, are reported in Table 1. Numbers in brackets are sample size

So what if we are wrong?

"Most models are wrong, but some are useful". (George Box)

"Most papers are wrong, but some are useful". (Me).

Solutions?

What can you as a researcher do?

- Forget about the journal title and its impact factor, share with your colleagues interesting research that you find (Twitter).
- Evaluating research: use the Sydney Brenner "trick" (Brenner, 1995)

  "we should remind ourselves that what matters absolutely is the scientific content of a paper and that nothing will substitute for either knowing it or reading it"
- Find a mutual network (reproducibility crowd)
- Investigate preprints . . .

Preprints

Rxivist

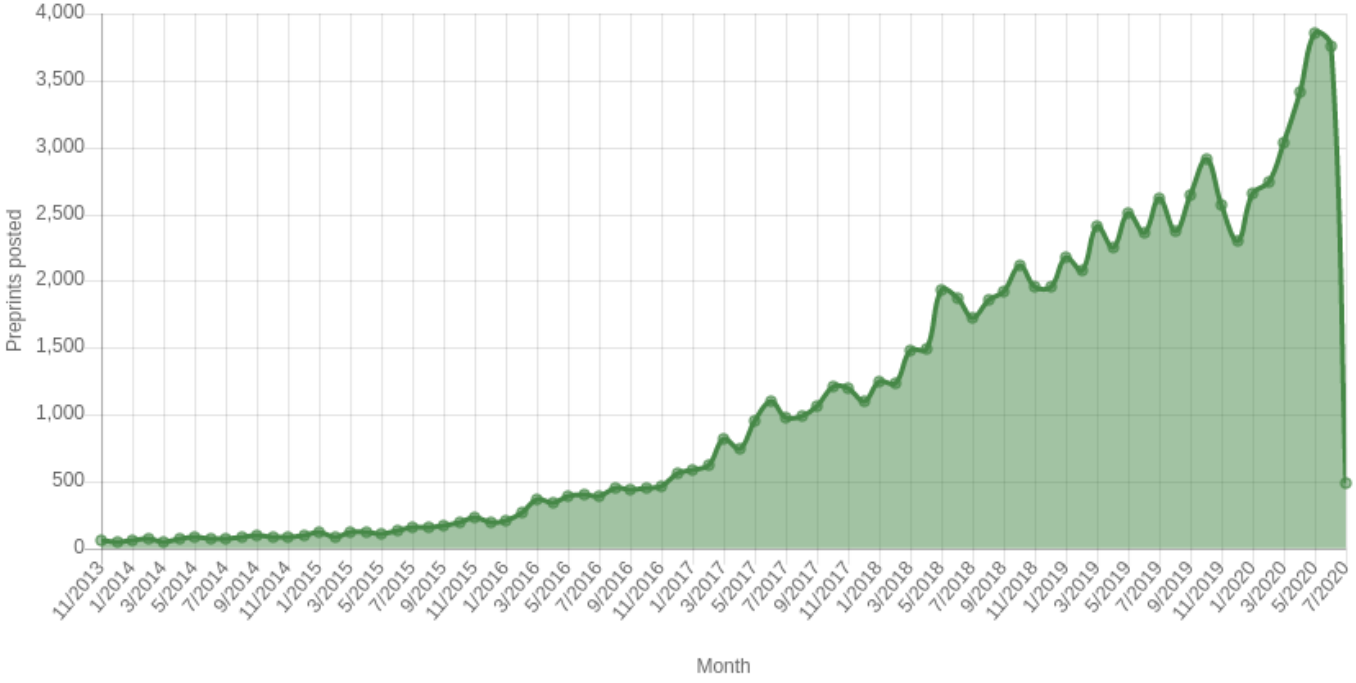
TRENDING OPEN SCIENCE

Rxivist combines preprints from bioRxiv with data from Twitter to help you **find the papers being discussed** in your field. Currently indexing **88,687 bioRxiv papers** from **380,283 authors**.

Site-wide metrics

The numbers below represent the metrics for all papers hosted on bioRxiv.org, based on our indexing of the website.

Monthly submissions, overall



Why preprint?

Advantages (https://www.plos.org/why-preprint):

1. Rapid dissemination of results
2. Establish priority
3. Increased attention/citations
4. Career advancement
5. Community
6. Unlimited / timely updates
7. Early feedback / error reduction
8. Some editors fish for papers on preprint servers
9. Preprints are valid research outputs for REF2021 Naomi Penfold.

Evaluate journal subscription model

Think about which journals you support, by sending papers and reviewing.

Support OA journals as authors and reviewers. (Ever asked a journal for permission to republish one of your own figures?)

Check out status of journal via Sherpa/Romeo service.

What can your institutions & funders do?

- sign up to DORA (Declaration of Research Assessment) (Wellcome Trust have solved this problem). Signing is easy, enforcing is hard.
- Give ECR a voice within institutions.
- Recognise all research artefacts, not just papers, as valuable outputs.
- Give academics more ownership of funds. "Do you really want to spend 5K on an APC, versus other uses in your lab?"
- Routes to OA need to be generate less admin for librarians / funders, not more. Green OA achieves that.
- Support diamond OA initiatives led by academics and academic societies.

Conclusions

- Many reasons to feel bad…
- But there are lots of reasons for optimism at new ways of doing science.
- Learn how to become an open scientist and use modern tools.
- Science improves one preprint submission at a time.
- Read more at http://bulliedintobadscience.org and sign up!